

Ryan Law

📞 (626) 624-6787 | ✉️ ryanlaw0329@gmail.com

EDUCATION

University of California, Irvine <i>Incoming Transfer, B.S. in Mechanical Engineering</i>	Expected Fall 2026 Irvine, CA
Mt. San Antonio College <i>Mechanical Engineering Transfer Preparation</i>	2024 – 2026 Walnut, CA
• Relevant Coursework: Statics, Dynamics, Mechanics of Materials, Properties of Materials, MATLAB, Engineering Graphics	

EXPERIENCE

UCLA California NanoSystems Institute (CNSI) <i>Microfabrication and Nanopatterning Programs</i>	Oct. 2025 – Dec. 2025 Los Angeles, CA
• Fabricated wafer-based microstructures using photolithography, thin-film deposition, wet etching, and electron-beam lithography in cleanroom environments	
• Optimized fabrication workflow by evaluating resist thickness, exposure conditions, and process sequencing to improve pattern quality and repeatability	
• Characterized fabricated features via microscopy and applied the findings to refine subsequent wafer iterations	
• Coordinated task division among team members during cleanroom processing to improve efficiency while maintaining contamination-control and safety standards	
Mt. San Antonio College VEX Robotics Team <i>VEX Robotics Design Lead</i>	Dec. 2024 – Mar. 2025 Walnut, CA
• Led mechanical design of a collegiate VEX robot for a 6-member team using SolidWorks and Fusion 360 for CAD, subsystem iteration, and fabrication planning, including drivetrain design with gears and other machine elements	
• Improved robot performance through 3D-printed component integration, geometric redesign, prototyping, and iterative testing	
• Managed subteam workflow, task allocation, and build schedules during competition preparation	
• Mentored newer members in CAD, fabrication, and system integration	
Mt. SAC UAV Team <i>Systems Engineer & 3D Design Engineer</i>	2nd Place Overall, 2026 C-UASC Walnut, CA
• Served as systems engineer and 3D design engineer for the Mt. SAC UAV Team, the highest-placing U.S. team at the California Unmanned Aerial Systems Competition (C-UASC), hosted by Cal State LA	
• Modeled the aircraft in CAD to support subsystem layout and competition-ready integration	
• Developed ground control software and mission planning tools to support autonomous flight operations during competition missions	

PROJECTS

RC Vehicle Aerodynamics Testing & Optimization	ESP32, SolidWorks, 3D Printing
• Designed and built an RC car from scratch using SolidWorks, 3D-printed components, and an ESP32-based control system	
• Programmed the ESP32 and integrated it with the vehicle's electronic drive system	
• Applied Reynolds number, pressure drag, and boundary-layer analysis to refine body geometry and improve vehicle stability and aerodynamic performance	
• Incorporated an internal airflow channel inspired by Formula 1 F-duct concepts to reduce drag and improve overall vehicle performance	
Folding Knife Design & Fabrication	Magnacut Steel, SolidWorks, Precision Fabrication
• Designed and hand-fabricated a folding knife from Magnacut steel, incorporating a back lock mechanism with tight component tolerances to ensure reliable locking and smooth operation	
• Evaluated material behavior, load paths, and component tolerances through prototyping and mechanical design analysis to support strength, durability, and reliability	
• Applied detail drawing and fabrication principles to translate CAD geometry into a functional, precision-fit mechanical assembly	
BroncoHacks Posture Device	Python, Raspberry Pi, Sensors
• Built a posture-monitoring prototype using a Raspberry Pi, distance sensor, breadboarded circuitry, and a 3D-printed enclosure	
• Developed a Python-based interface to support device operation, sensor feedback, and user interaction	
• Integrated hardware and software under a 48-hour hackathon timeline to address a real-world posture problem	
Raspberry Pi Sense HAT Controller	MATLAB, Raspberry Pi, Sense HAT
• Developed a MATLAB-based control interface for an 8x8 LED matrix using joystick and accelerometer inputs	
• Integrated sensor processing, conditional logic, and display control in an embedded mechatronic system	

SKILLS

Aerospace & Structures: Mechanics of materials, dynamics, structural and stress analysis, propulsion-relevant motion analysis
CAD/Design: SolidWorks, Fusion 360, engineering drawings, GD&T
Fabrication & Manufacturing: 3D Printing, Laser Cutting, hand fabrication, component tolerancing, material selection
Programming: MATLAB, Python, C++, Arduino